EE-170L Computer Programming Lab

Laboratory No 01

Installation of DEV-C++ IDE and Execution of First C++ Program

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Lab Objectives

After completing this lab, students will be able to:

1. Download and install DEV-C++ IDE
2. Open and explore the DEV-C++ environment
3. Create a new C++ source file
4. Write, compile, and run a basic C++ program
5. Understand the output window and common buttons

Task-01

1. **Download and install the latest version of the IDE:**
   - Link to download: https://sourceforge.net/projects/orwelldevcpp/

     Or

   - https://sourceforge.net/projects/embarcadero-devcpp/

2. **Create a New C++ Program**
   - **i. Click File → New → Source File**
   - **ii. A blank editor window will open**
   - iii. Click **File → Save As**
   - iv. File name: lab1_hello.cpp
3. **Write First C++ Program**

```
#include <iostream>
using namespace std;

int main() {
   cout << "Hello World!";
   return 0;
}
```

4. **Compile and Run the Program**
   - **i. Click Execute → Compile**
   - **ii. Click Execute → Run**

Optional Tasks:

Task-01

Installation of VS Code IDE and Running First C++ Program

Objectives

Students will learn how to:

1. Install Visual Studio Code (VS Code)
2. Install a C/C++ compiler (MinGW) and configure it
3. Add required extensions in VS Code
4. Write, build/compile, and run a "Hello World!" C++ program

(Sheraz sb lab-01)

Helping material:

https://www.youtube.com/watch?v=1PBD5qFWdq8

https://www.youtube.com/watch?v=DMWD7wfhgNY

Task-02: Creating GitHub Account and Uploading C++ Lab Files

Objectives

After completing this lab, students will be able to:

i. Create a GitHub account
ii. Create a new repository
iii. Upload lab files (.cpp)
iv. Share repository link with instructor

Step-1: Open GitHub Website

1. Open **Google Chrome**
2. Go to the official website:
   https://github.com

Step-2: Sign Up for a GitHub Account

1. Click **Sign up**
2. Enter:
   - **Email address**
   - **Password**
   - **Username**

3. Click **Continue**

Step-3: Choose Account Preferences

1. Select **Free plan**

Step-4: Create Your First Repository (Lab Folder)

1. Click **New** or **+ → New repository**
2. Fill details:
    - **Repository name: Cpp-Lab-Tasks**
    - Description (optional): C++ Programming Lab Tasks – Semester 2

- Select **Public**

Tick **Add a README file**

3 Click **Create repository**

Step-5: Upload First Lab File (Hello World Program)

Method-1: Upload via Browser (Recommended for Beginners)

1. Inside your repository, click **Add file → Upload files**
2. Click **Choose your files**
3. Select: lab1_hello.cpp
4. Scroll down and click **Commit changes**

Or

Step-6: Create Folder for Each Lab (Good Practice)

1. Click **Add file → Create new file**
2. In file name, type:
3. Lab-1/lab1_hello.cpp
4. Paste your C++ code
5. Click **Commit changes**

Step-7: Upload VS Code / DEV-C++ Lab Files

For future labs:

- Lab-2 → Lab-2/input_output.cpp
- Lab-3 → Lab-3/loops.cpp
- Lab-4 → Lab-4/functions.cpp

Keep **one folder per lab**

Step-8: Copy Repository Link for Submission

1. Click **Code → Copy HTTPS link**
   OR
2. Copy browser URL directly

Example:

https://github.com/username/Cpp-Lab-Tasks

Submit the link to the instructor.

Task:

1. Create a GitHub account
2. Create a repository named Cpp-Lab-Tasks
3. Upload: Lab-1/lab1_hello.cpp
4. Submit repository link

The End

Explore the following resources

1. Introduction to stack overflow
   - https://stackoverflow.com/
   - Search the answer for most common programming questions
   - Search for error messages to see what cause them
   - Ask programming questions to get Answer from programmer worldwide
   - Gain valuable insights about programming in general
2. Geeks For Geeks
   - https://www.geeksforgeeks.org/
3. W3School
   - https://www.w3schools.com/CPP/default.asp